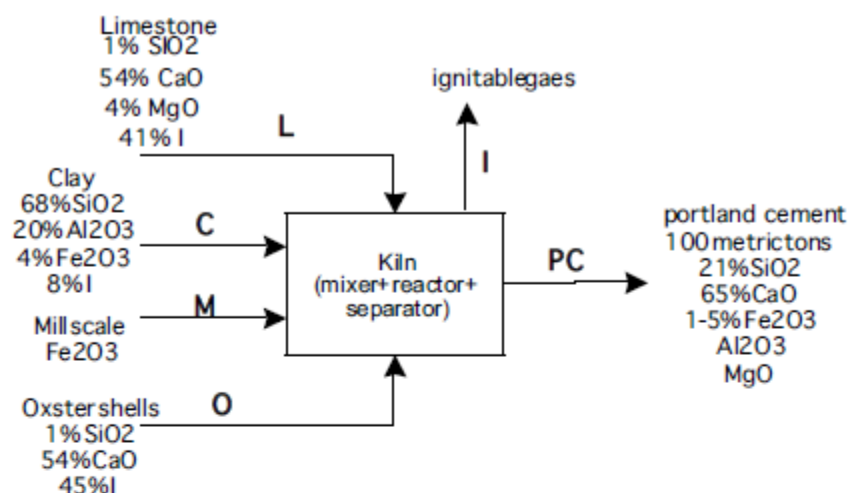


P2.67

We'll combine mixing and heating to drive off gases in a single flow diagram:



The streams are labeled as L, C, M and O for limestone, clay, mill scale and oyster shells, respectively. There are 6 components. All units are in tons or wt%, so no conversions are needed.

DOF analysis:

	Variables	Constraints
Stream	18	
System	0	
Specified flows		1
Specified compositions		10
Specified system performance		0
Material balance		6

$\text{DOF} = 18 - 17 = 1$. We have an underspecified problem, but we did not count the constraint that Fe_2O_3 must be between 1-5%. Within these limits, our objective is to minimize the total cost.

It is most convenient to work in mass fractions and total quantities of each stream, because the compositions are known but the quantities are unknown. Combining stream composition specifications with material balance equations, we find:

$$\text{SiO}_2: 0.01F_L + 0.68F_C + 0.01F_O = 0.21(100) = 21 \text{ tons}$$

$$\text{CaO}: 0.54F_L + 0.54F_O = 0.65(100) = 65 \text{ tons}$$

Although there are only 2 equations and 3 unknowns, we can partially solve for some useful information:

$$F_L + F_O = 120.4 \text{ tons}$$

$$F_C = 29.1 \text{ tons}$$

We have an additional relationship from the Fe_2O_3 balance:

$$1 \leq 0.04F_L + F_M \leq 5 \text{ tons}$$

and, from the total mass balance:

$$0.59F_L + 0.92F_C + F_M + 0.55F_O = 100 \text{ tons}$$

Finally, we want to find the quantities that provide the minimum cost,

$$\text{Cost}(\$) = 88F_L + 35F_C + 60F_O$$

Although the limestone is more expensive than the oyster shells, it also has more MgO in it, which reduces the amount of (free) mill scale we might be able to blend in. Thus, the cheapest combination is not immediately obvious. We set up a spreadsheet to analyze the problem.

Iron content	Fc	Fm (from Fe_2O_3 balance)	FL (from total balance)	FO (from CaO balance)	cost (\$)	
1	29.1	-0.164	179.3	-58.9	--	--
2	29.1	0.836	154.3	-33.9	--	--
3	29.1	1.836	129.3	-8.9	--	--
4	29.1	2.836	104.3	16.1	11163	
5	29.1	3.836	79.3	41.1	10463	

Values below 3% iron are not physically possible. The cheapest combination gives 5% iron, with 29.1 tons clay, 3.8 tons mill scale, 79.3 tons limestone, and 41.1 tons oyster shells.

P3.21

The system is the body, and caffeine is the component. DOF analysis is similar to P3.19. First we find an expression for the flow rate of caffeine out:

$$\frac{dm_{c,sys}}{dt} = -\dot{m}_{c,out} = -0.116m_{c,sys}$$

where the rate is given in mg caffeine/h.

Rearranging gives

$$\frac{dm_{c,sys}}{m_{c,sys}} = -0.116dt \quad (\text{Eq. 1})$$

A general equation describing the caffeine in the body at any time is obtained by integrating with the initial condition that $m_{c,sys} = m_{c,sys,0}$ at $t = 0$:

$$m_{c,sys} = m_{c,sys,0} \exp(-0.116t) \quad (\text{Eq. 2})$$

First we want to find out how long it takes for body caffeine content to drop to 100 mg after drinking a 200 mg cup. We integrate Eq. 1 within specific limits:

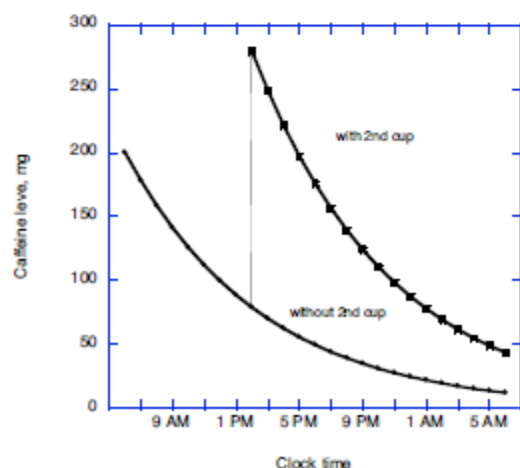
$$\int_{200}^{100} \frac{dm_{c,sys}}{m_{c,sys}} = \int_0^{t_f} -0.116dt$$

$$\ln\left(\frac{100}{200}\right) = -0.116t_f$$

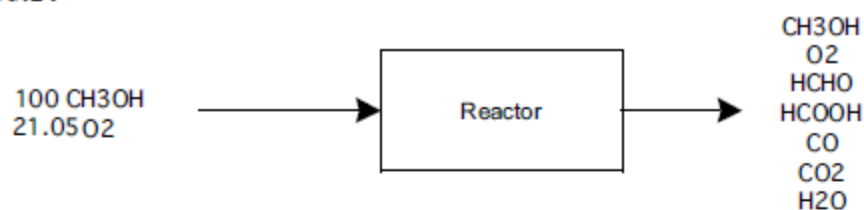
$$t_f = 5.97$$

It takes about 6 h for the caffeine level to drop from 200 to 100 mg.

To plot the caffeine in the body as a function of time, with two cups at 6 AM and at 2 PM, we divide the problem into two time intervals. In the first time interval, which lasts 6 hr, we assume that $m_{c,sys,0} = 200$ mg. In the second interval (2 PM to 11 PM, or 9 h), we assume that $m_{c,sys,0} = 200 \text{ mg} + m_{c,sys,f}$ from the first time interval. We use Eq. 2 to calculate $m_{c,sys}$ at any time.



P3.24



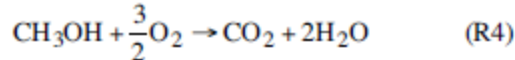
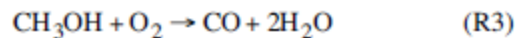
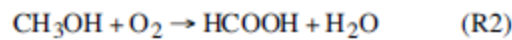
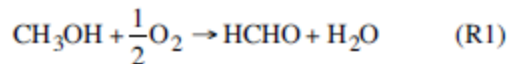
DOF analysis:

	No. of variables		No. of constraints
Stream variables	9	Specified flow	2
System variables	4 (reactions, because there are 7 compounds and 3 elements)	Specified compositions	0
		Specified performance	2 (fractional conversions)
		Material balances	7
Total	13		11

DOF = 13 - 11 = 2, underspecified??

Let M = methanol, O = O₂, F = formaldehyde, FA = formic acid, CM = carbon monoxide, CD = carbon dioxide, W = water. The system is the reactor, and we'll assume it is steady state. All flows in kgmol/h.

A set of 4 independent balanced chemical reactions that describe this system is:



From the reactor performance specifications:

$$\dot{n}_{M,out} = (1 - f_{C,M}) \dot{n}_{M,in} = (1 - 0.4)100 = 60$$

$$\dot{n}_{O,out} = (1 - f_{C,O}) \dot{n}_{O,in} = (1 - 0.95)21.05 = 1.05$$

From material balance equations:

$$\dot{n}_{M,out} = \dot{n}_{M,in} + \sum v_{ik} \dot{\xi}_k = 60 = 100 - (\dot{\xi}_1 + \dot{\xi}_2 + \dot{\xi}_3 + \dot{\xi}_4)$$

$$\dot{\xi}_1 + \dot{\xi}_2 + \dot{\xi}_3 + \dot{\xi}_4 = 40$$

$$\dot{n}_{O,out} = \dot{n}_{O,in} + \sum v_{ik} \dot{\xi}_k = 1.05 = 21.05 - \left(\frac{1}{2} \dot{\xi}_1 + \dot{\xi}_2 + \dot{\xi}_3 + \frac{3}{2} \dot{\xi}_4 \right)$$

$$\frac{1}{2} \dot{\xi}_1 + \dot{\xi}_2 + \dot{\xi}_3 + \frac{3}{2} \dot{\xi}_4 = 20$$

We have 2 equations in 4 unknowns. Can we get anywhere? Perhaps. Suppose we try subtracting the two equations we derived from material balances;

$$\dot{\xi}_1 + \dot{\xi}_2 + \dot{\xi}_3 + \dot{\xi}_4 - \left(\frac{1}{2} \dot{\xi}_1 + \dot{\xi}_2 + \dot{\xi}_3 + \frac{3}{2} \dot{\xi}_4 \right) = 40 - 20$$

$$\frac{1}{2} \dot{\xi}_1 - \frac{1}{2} \dot{\xi}_4 = 20$$

$$\dot{\xi}_1 - \dot{\xi}_4 = 40$$

Now we note that all extents of reaction must be positive, because only methanol and oxygen are fed to the reactor. This is an additional constraint on the solution that was not explicitly considered in the DOF analysis. Although there are an infinite number of mathematical solutions, there is only one physically reasonable solution:

$$\dot{\xi}_1 = 40, \dot{\xi}_4 = \dot{\xi}_3 = \dot{\xi}_2 = 0$$

The material balances on the other components become:

$$\dot{n}_{F,out} = \sum v_{ik} \dot{\xi}_k = \dot{\xi}_1 = 40$$

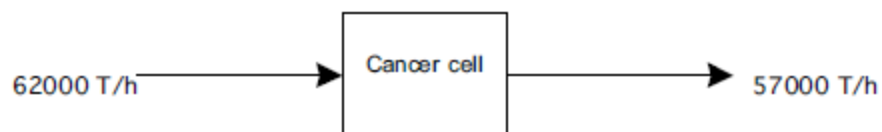
$$\dot{n}_{W,out} = \sum v_{ik} \dot{\xi}_k = \dot{\xi}_1 = 40$$

$$\dot{n}_{FA,out} = \dot{n}_{CM,out} = \dot{n}_{CD,out} = 0$$

The reactor outlet flow is 60 kgmol/h methanol, 40 kgmol/h formaldehyde, 40 kgmol/h water, and 1.05 kgmol/h oxygen. In terms of mol% and total flow: 141.05 kgmol/h at 42.5% methanol, 28.4% each formaldehyde and water, and 0.7% oxygen.

P3.29

The immunotoxin (T) is the component of interest, and the cancer cell is the system. Inside the system, both accumulation and reaction occur



DOF analysis:

	No. of variables		No. of constraints
Stream variables	2 (T in, T out)	Specified flow	2
System variables	2 (accumulation, reaction)	Specified compositions	0
		Specified performance	1 (reaction rate)
		Material balances	1
Total	4		4

$$\text{DOF} = 4 - 4 = 0.$$

An integral balance is appropriate, since we are interested in accumulation after a finite time interval.

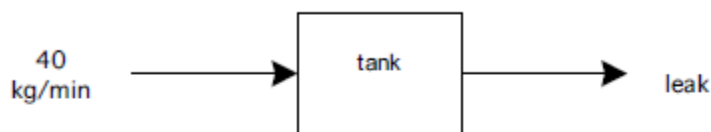
$$m_{T,sys,f} - 0 = \int_0^8 62000 dt - \int_0^8 57000 dt + \int_0^8 -2700(1 - e^{-0.3t}) dt$$

$$m_{T,sys,f} - 0 = 40000 - \left[2700(8) + \frac{2700}{0.3} (e^{-0.3(8)} - e^0) \right] = 40000 - 21600 + 816 - 9000 = 26,580$$

The product comes close but doesn't quite achieve the goal of 30,000 immunotoxin molecules accumulated in the cell over 8 h. If you believe that the technology could be improved a bit, though, it looks like a pretty good investment.

P3.41

The system (the tank) is unsteady state. Initially, there is a flow into the tank but none out. Later, there is both flow in and flow out, and the rate of the flow out changes with time, as the system fills.



We can either do a balance on compound X, or on total mass, because the composition doesn't change. We'll use compound X.

From $t = 0$ to $t = t_{leak}$, there is no flow out, and the tank is initially empty. At the point of

the corroded spot, the tank volume is $\frac{\pi(1\text{ m}^2)}{4} \times 1\text{ m} = 0.785\text{ m}^3$. Given the density, this is a mass of 785 kg (78.5 kg compound X). From the integral mass balance on compound X, we find:

$$m_{X,sys,f} - m_{X,sys,0} = \int_0^{t_{leak}} \dot{m}_{X,in} dt$$

$$78.5 = 4.0(t_{leak} - 0)$$

$$t_{leak} = 19.6\text{ min}$$

After 19.6 min, there is an outlet flow that scales with the height of the fluid in the tank h (in meters). The flow rate of compound X out of the tank (kg/min) is given as:

$$\dot{m}_{X,out} = 0.4\sqrt{h-1}$$

The height of the mass in the tank depends on the flow rate in and out! We can write the differential mass balance equation, taking advantage of the known density and the dimensions of the cylindrical tank:

$$\frac{dm_{X,sys}}{dt} = (0.1 \text{ kg X/kg}) \left(\frac{\pi}{4} \text{ m}^2 \right) \times 1000 \text{ kg/m}^3 \times \frac{dh}{dt} = \dot{m}_{X,in} - \dot{m}_{X,out} = 4 - 0.4\sqrt{h-1}$$

Simplifying, we get:

$$78.5 \frac{dh}{dt} = (4 - 0.4\sqrt{h-1})$$

Now we can rearrange, and write as an integral, from $t = 19.6$ min to $t = 40$ min, and from $h = 1$ m to $h = h_f$:

$$78.5 \int_1^{h_f} \frac{dh}{(4 - 0.4\sqrt{h-1})} = \int_{19.6}^{40} dt = 20.4 \text{ min}$$

We have a couple of options. If we remember our calculus, we can derive an analytical solution. If not, we can use the numerical integration function on our calculator (or an equation solver), and just guess values of h_f till we find one that is consistent with the above equation. We know that h_f has to be greater than 1 m and less than 3 m.

Guess: 2 m: integral = 0.268

Guess: 1.95 m: integral = 0.254

Guess: 1.96 m: integral = 0.256

Guess: 1.975 m: integral = 0.261 (close enough)

The total amount of compound X pumped into the tank from 19.6 min to 40 min is simply $(0.1 \text{ kg X/kg soln})(40 \text{ kg soln/min})(40 - 19.6 \text{ min}) = 81.6 \text{ kg}$. The total amount of compound X added to the tank from 1 m high to 1.975 m high is

$$(1.975 - 1 \text{ m}) \left(\frac{\pi}{4} \right) (1 \text{ m})^2 (1000 \text{ kg/m}^3) (0.1 \text{ kg X/kg soln}) = 76.5 \text{ kg}$$

Therefore, the amount of compound X on the floor is $81.6 - 76.6$ or about 5.1 kg. (You will get slightly different answers depending on the accuracy to which you solved the integral.)

(Another way to look at this follows: If there were no leak, we'd have

$$\frac{dm_{X,sys}}{dt} = (0.1 \text{ kg X/kg}) \left(\frac{\pi}{4} \text{ m}^2 \right) \times 1000 \text{ kg/m}^3 \times \frac{dh}{dt} = \dot{m}_{X,in} - \dot{m}_{X,out} = 4 - 0 = 4 \text{ kg/min}$$

or, rearranging and integrating,

$$19.63 \int_1^{h_f} dh = \int_{19.6}^{40} dt = 20.4 \text{ min}$$

Solving, we find $h_f = 2.04$. The difference in height with and without the leak corresponds to the mass of material leaked onto the floor. We calculate this as

$$(2.04 - 1.975 \text{ m}) \left(\frac{\pi}{4} \right) (1 \text{ m})^2 (1000 \text{ kg/m}^3) (0.1 \text{ kg X/kg soln}) = 5.1 \text{ kg}$$